



Day 15 Classical Baseline & Mentor feedback

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Classical Baseline Review

Objective:

- Review classical underwater image enhancement methods.
- Compare their performance using quantitative metrics.
- Analyse edge preservation and visual quality.
- Select a strong classical reference baseline.
- Confirm the direction toward a learning-based enhancement approach.

Methods Evaluated:

- CLAHE
- Gamma Correction
- White Balance

Evaluation Metrics:

- PSNR
- SSIM
- Edge F1

Classical Baseline Results

<u>Method</u>	<u>PSNR</u>	<u>SSIM</u>	<u>Edge F1</u>
CLAHE	10.99	0.4346	0.0572
Gamma Correction	10.93	0.4821	0.0416
White Balance	10.44	0.4611	0.0416

Observations:

- CLAHE achieved the highest PSNR: 10.99.
- Gamma Correction achieved the highest SSIM: 0.4821.
- CLAHE achieved the highest Edge F1: 0.0572.
- White Balance achieved the lowest PSNR.
- No method was best on every metric.

CLAHE is the strongest overall classical reference based on PSNR and Edge F1.

Visual Findings

Visual Observations:

Input

- Very dark and low visibility.
- Important details are difficult to identify.

Target

- Brighter and clearer.
- Better visibility of objects and scene details.

White Balance

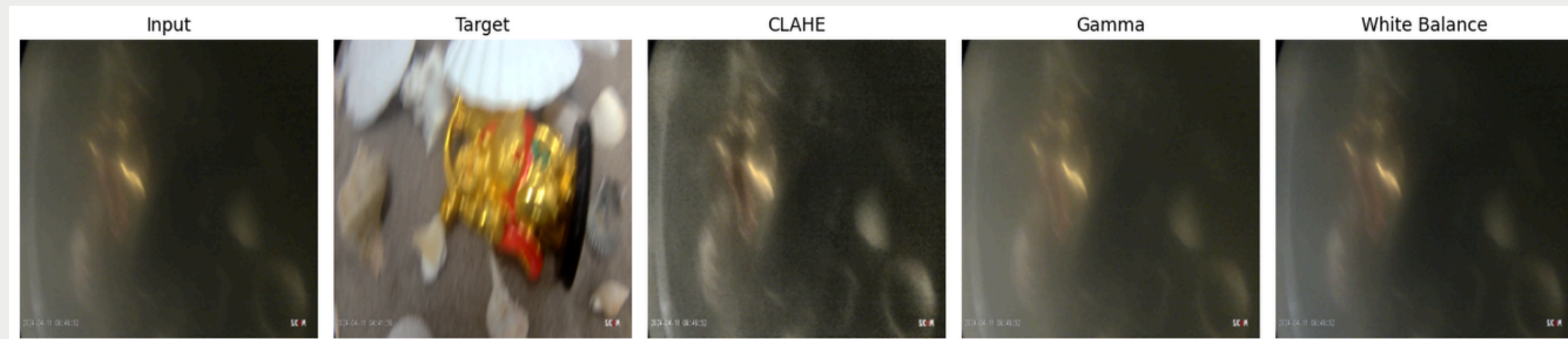
- Provides limited visual improvement.
- Less effective for severely dark scenes.

Gamma Correction

- Provides some brightness improvement.
- Limited recovery of hidden details.

CLAHE

- Improves local contrast.
- Makes some details more visible.





Fair Evaluation and Baseline Decision

Evaluation Metric Definitions

PSNR

- Measures pixel-level similarity between output and target.
- Higher value indicates better similarity.

SSIM

- Measures structural similarity between output and target.
- Considers brightness, contrast and structure.
- Higher value is better.

Edge F1

- Measures similarity between edges in the enhanced output and target.
- Higher value indicates better edge agreement.

Fair Comparison

- All three methods were evaluated using:
- Same input images
- Same corresponding target images
- Same evaluation procedure
- Same metrics

Baseline Decision

CLAHE → Selected as the strongest
classical reference

Reason:

- Highest PSNR: 10.99
- Highest Edge F1: 0.0572
- Provides noticeable local contrast improvement.

Mentor Review

Sir you can write your feedback in Readme and can you answer this questions so we can move further in any of the Learning based Enhancement Techniques.

Questions

1. Sir are PSNR, SSIM and Edge F1 suitable for this project?
2. Sir is CLAHE acceptable as the classical reference baseline?
3. Should we need to add any classical method or metric?
4. Is the current input-target comparison fair?
5. Are there any preprocessing or evaluation corrections?
6. Can we proceed to the learning-based enhancement stage?



Thank
You

